

Annual Progress Report – Kokos-me-rpgle-smeuperp Project

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1. Key Progress by Product Area

1.1. Core Functionality and Interoperability

- **FUN.DIFF Service Enhancements:** The `MURUNT_01` service has been significantly improved to render errors with stack-traces directly in HTML, making debugging more efficient. This service now also includes a new `Errors` column in its output, indicating specific errors for "actual" executions, with "Number of differences" and "Differences" columns set to `-1` and blank respectively when errors occur.
- **Parameter Logging Improvements:** The `ParamsLogger` utility class has been enhanced to "reduce" the display of Data Structure (DS) fields, providing a more concise and readable output.
- **£G90 Implementation:** Full implementation of the `£G90` program, including `LET` function with `SETLL` and `READ` methods. It now supports options for "only tag" (considering only lines with a tag), "include" (incorporating files based on tags and File/Member specifications), and "ignore blank" lines. Additionally, it can concatenate lines if the last character is `+`.
- **Date Handling (£DA8):** Separated the parsing logic for special dates (like `&OGI00`) from `£DA8Service` into a new `SpecialDateParser` class, which can also be useful for `£G43`. `£DAT8` was improved to cover the `SM` action for incrementing months and the `FGM` code for forcing a date to a specific day of the month.
- **£RITES Improvements:** Enhanced `£RITES` by caching `ResultSet` for `readNext` and adding `ORDER BY` and a second `WHERE` condition for position and `positionAndRead` methods, leading to significant performance improvements. Fixed an issue where `moveCursorToElement` in `£RITES` was throwing errors prematurely, ensuring correct cursor positioning and element finding.

1.2. Robustness and Stability

- **G90 Logic Refinement:** Tweaked `G90` logic to properly handle error indicators and fixed the debug logger that was outputting incorrect values.
- **JDBC Driver Stability:** Added a new caching mechanism for JDBC SELECT query results implemented as a custom proxy driver (`CacheJdbcDriver`), which includes connection parsing, cache mode handling (memory or file-based), and backend driver loading, enhancing

stability for database operations.

- **Path Sanitization:** Implemented path sanitization for `G90Service`, improving robustness when handling file paths.
- **Doped Program Test Generation:** Introduced a mechanism to auto-generate test suites for doped programs directly from AS400 specifications, significantly increasing test coverage and identifying issues early.
- **Error Handling in `BEAR10`:** Removed exceptions in `BEAR10` and replaced them with indicator settings, preventing crashes in certain scenarios.

1.3. Observability and Diagnostics

- **HTML Stack-trace for Errors:** `MURUNT_01` service now provides detailed stack-traces in HTML for `FUN.DIFF` when an error occurs, greatly aiding in error diagnosis.
- **Error Column in `MURUNT_01`:** A new `Errors` column in `MURUNT_01` service output explicitly indicates execution errors.
- **Verbose Test Output:** Removed the `--quiet` flag from `mvn test` in the GitHub Actions workflow, enabling more detailed output during test execution for better diagnostics.
- **Improved Mock Logging:** Enhanced `MockedKokosProgram` to log parameter values for both input and output, providing more visibility into mocked program executions.

1.4. Performance Optimization

- **`ERITES` Caching and Query Optimization:** Implemented `ResultSet` caching and optimized SQL queries (`ORDER BY`, `WHERE` conditions) in `ERITES` to reduce execution time, demonstrating significant performance gains.
- **`BEAR10` Query Performance:** Refactored `BEAR10` to improve query performance by caching results on position operations.

1.5. Maintenance and Technological Upgrades

- **Documentation Enhancement:** Refined documentation for `Fun` comparison service methods (`MURUNT_01`), including parameter descriptions and consistent annotations, improving clarity and usability for developers. Comprehensive documentation for generating Markdown summaries of pull requests was also added, outlining processes, grouping rules, and output formats.
- **Code Structure Standardization:** Migrated `BEINZ0` to the new standard structure. Refactored mocked programs to a new standardized structure, replacing `MockedKokosProgram` with `MockedService` at the service level. Aligned existing `COPY` package names to new standards. Refactored `BE675G`, `BE6CK0CL`, `BE649G`, and `BE654G` to align with new standards and separate library management logic.

- **CI/CD Pipeline Adjustment:** Removed the step for updating containers on the Kokos staging environment from `develop-release.yml`, streamlining the CI/CD pipeline.
- **Dependency/Version Bumps:** Updated `pom.xml` to include the H2 database dependency for testing. Updated dependencies and centralized JUnit version.
- **New Doped Programs and Mocks:** Added doped implementations or mocks for `BEG43E`, `BEGCK02`, `BEAR10X`, `BEG91G`, `JAJ14G`, `BEWK12CL`, `BEG49G`, `BEG90GW`, `BEG00A`, `BENS00`, `JAJ180`, `BEGGP0`, `BEWK20CL`, `BECRI0` and `BECRI1`.
- **Refactoring `HttpClient`:** The `LaunchFun` class, used for executing FUNs, addresses a timeout behavior in `HttpClient` that was not always considered.

2. Number of Pull Requests and Contributors

- **Total Number of Merged Pull Requests:** 50
- **Unique Contributors:** 6 (davidepalladino-apuliasoft, lanarimarco, dom-apuliasoft, mattiabonardi, GaetanoDiNatale, mossini-smeup)
- **Breakdown:**
 - davidepalladino-apuliasoft: 24 PRs
 - dom-apuliasoft: 13 PRs
 - lanarimarco: 11 PRs
 - mattiabonardi: 4 PRs
 - GaetanoDiNatale: 3 PRs
 - mossini-smeup: 1 PR